

# Appendix C: IEEE 830 Template

**Software Requirements Specification**

**for**

**Smart Parking System**

**Version 1.0**

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## **Table of Contents**

**1.Introduction Page 1-2**

**2. Overall Description page 2-4**

## **Revision History**

| Name               | Date       | Reason for Changes | Version |
|--------------------|------------|--------------------|---------|
| Ismail Eyup Gocmen | 14.04.2026 |                    | 1,0     |

## **1. Introduction**

### **1.1. Purpose**

This document explains the specifics of the smart parking system. The purpose of this project is to create a smart parking system that allows users to locate, reserve, and pay for parking spaces efficient. It defines the system's functionalities, constraints, and interfaces for development and evaluation.

## 1.2. Document Conventions

This document follows the IEEE 830 Software Requirements Specification standard and uses consistent terminology and formatting conventions to ensure clarity and precision. All mandatory requirements are expressed using the keyword “**shall**”, indicating a mandatory system. Optional or non-mandatory features may use terms such as “may” or “can”.

Functional requirements are labeled using the format Req-x, where x represents a unique identifier. Non-functional requirements are labeled as NF-x and are categorized based on quality attributes such as performance, security, and reliability.

## 1.3. Intended Audience and Reading Suggestions

This document is intended for developers, testers who will test the system, stakeholders who want to know more about their investment, and users that want to know more about how the system works.

**Developer:** Developers who intend to work on the system whether to improve, modify, or extend the system must consult the document to understand the requirements and ensure consistency across all development phases.

**Stakeholders:** Stakeholders and including customers review this document to know if the requirements aligns with their requirements and goals. They use this specification to ensure that the system fulfills its intended purpose and delivers the required functionality.

**Tester:** The tester uses this document to test if the implemented system satisfies all specified requirements. It serves as a reference for designing test cases and ensuring that the system behavior matches the defined requirements.

## 1.4. Product Scope

The Smart Parking System is designed to assist drivers in locating, reserving, and paying for parking spaces in an efficient and manner. The system provides real-time information about parking availability and enables users to reserve spots in advance, reducing the time spent searching for parking. The system also supports administrative functionality for managing parking spaces, including adding, updating, and removing parking data. Additionally, it integrates with an external payment system to handle secure transactions.

## 1.5. References

No references